



Exploring the mediating roles of depression and cognitive function in the association between sarcopenia and frailty: A Cox survival analysis approach

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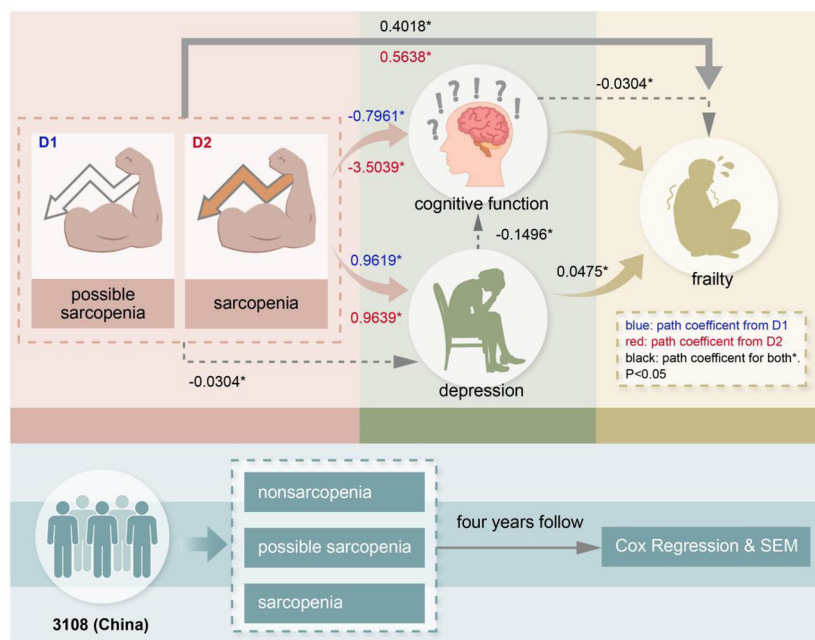
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HIGHLIGHTS

- Baseline sarcopenia, depression, and cognitive function were significantly associated with the onset of new-onset frailty.
- The associations between depression, cognitive function, and frailty vary across different sarcopenia statuses.
- Sarcopenia is not only associated with frailty through depression and cognitive function separately but also affects frailty via the chain-mediated effects of depression and cognitive function.

GRAPHICAL ABSTRACT



ARTICLE INFO

Article history:

Received 20 October 2024

Revised 14 December 2024

ABSTRACT

Background: Despite earlier research indicating a potential link between the development of sarcopenia and an elevated risk of frailty, the lack of comprehensive prospective data on the correlation between sarcopenia and frailty incidence leaves open the question of whether depression and cognitive function mediate this association.

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<https://doi.org/10.1016/j.jare.2024.12.021>

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Accepted 16 December 2024
Available online 17 December 2024

Keywords:

Sarcopenia
Depression
Cognitive function
Frailty

Objective: The principal aim of the current investigation was to evaluate the intricate interplay among sarcopenia, depression, and cognitive function collectively influence the risk of developing frailty.

Methods: The participants included in this study were obtained from three waves of the China Health and Retirement Longitudinal Study (CHARLS), which collectively encompassed a total of 3,108 participants. To examine the interrelationships among sarcopenia, depression, cognitive function, and the incidence of frailty, we employed Cox regression models along with structural equation modelling, while making necessary adjustments for baseline demographic characteristics and various lifestyle factors.

Results: During a 4-year follow-up, we documented 753 frailty events. Compared to those with nonsarcopenia, those with possible sarcopenia and sarcopenia presented risk ratios for frailty events of 1.354 (95 % CI: 1.156, 1.586) and 1.514 (95 % CI: 1.203, 1.907), respectively. Stratified analyses by different statuses of sarcopenia further revealed that the significant effect of depression on frailty was present across all groups (nonsarcopenia, possible sarcopenia and sarcopenia), whereas the effect of cognitive function on frailty was limited to the non-sarcopenia and possible sarcopenia groups. Mediation analysis showed that sarcopenia was correlated not only with frailty through depression and cognitive function separately but also through a chain-mediated effect of depression and cognitive function together.

Conclusions: Sarcopenia is associated with frailty, depression and cognitive function playing partial, mediating roles. Frailty's susceptibility to depression and cognitive function differs based on sarcopenia status. Therefore, comprehensive interventions that include sarcopenia screening, interventions, improvements in depression, the promotion of mental health, and delays in cognitive decline will be more effective in preventing and delaying frailty. This effectiveness is particularly relevant for middle-aged and older adults who reside in China.

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Introduction

In the context of a globally aging population, a prevalent geriatric syndrome referred to as frailty is surfacing as a significant health burden[1,2]. This frailty represents a multidimensional physiological state involving declines in physical, immune, and metabolic functions[3], thereby elevating the susceptibility of older adults to diverse negative health outcomes and health-related incidents, such as falls, hospitalizations, heightened mortality rates, and diminished quality of life[1]. Globally, the incidence of frailty markedly escalates as individuals grow older. According to research employing the National Health and Nutrition Examination Survey (NHANES) in the United States, approximately 5–10 % of individuals who are 65 years of age and older exhibit frailty, whereas the prevalence of prefrailty reaches levels approaching 40–60 % [4]. A large multinational cohort study demonstrated notable variations in the prevalence of frailty across various nations, but the overall trend parallels closely with the findings documented in the United States[5]. Although the incidence of frailty does indeed increase with age, it is viewed as a pronounced manifestation of the typical aging process, suggesting that frailty may be preventable[1]. More importantly, an increasing body of evidence indicates that frailty can be reversed through appropriate interventions[6,7]. Therefore, identifying modifiable factors that could aid in preventing and slowing the progression of frailty is crucial.

Sarcopenia, a condition marked by the progressive deterioration of muscle mass, strength, and functionality as one ages, significantly impairs the quality of life and independence of elderly individuals[8]. Based on the definitions provided by the International Working Group on Sarcopenia (IWGS), sarcopenia involves reduced muscle mass, diminished muscle strength, and/or impaired physical performance[9]. Insights from a comprehensive meta-analysis indicated that the prevalence of sarcopenia varies between 10 % and 27 %, with severe sarcopenia affecting 2 % to 9 % of the population[10]. These figures are anticipated to steadily increase as the global population continues to age[11]. Numerous studies have linked sarcopenia to various negative health outcomes, including functional decline, physical disability, falls, prolonged hospital stays, and frailty in older adults[12–14]. Furthermore, sarcopenia not only affects physiological functions but is also closely associ-

ated with mental health and cognitive function. According to Chang et al., sarcopenia is independently linked to depression, which could directly lead to depressive symptoms[15]. Sarcopenia may influence mood states indirectly by disrupting the balance of the neuroendocrine system, particularly through the modulation of inflammatory responses in muscle tissue[16]. In terms of cognitive function, sarcopenia may negatively impact cognition by affecting cerebral blood supply and neuroplasticity[17]. Additionally, sarcopenia is known to increase the levels of inflammatory markers within the body, including interleukin-6 (IL-6) and tumour necrosis factor- α (TNF- α), which are associated with cognitive decline[17,18].

Previous studies have indicated that the existence of sarcopenia considerably heightens the likelihood of individuals developing frailty. Research conducted by Cruz-Jentoft et al. in 2010 highlighted that sarcopenia represents a multifaceted condition characterized by numerous contributing risk elements, which increase the risk of frailty through intricate pathways[14]. As proposed by Fried et al., sarcopenia occupies a vital pathogenic function in the progression of frailty and is a key factor in explaining many aspects of the frail state[3]. However, these studies have not explored the factors that might explain the link between sarcopenia and frailty. Current evidence suggests that sarcopenia significantly increases the likelihood of depression and cognitive decline[16], and both depression and cognitive decline are recognized as significant predictors that contribute to a heightened likelihood of experiencing frailty in older individuals[19]. Specifically, depression can lead to reduced physical activity[20], malnutrition[21], and impaired social participation[22], all of which are known risk factors for frailty[3,23,24]. Simultaneously, according to Beck's cognitive model of depression[25], once activated by stressors, depressive schemas can influence the processing of incoming information, leading to specific impairments in attention, interpretation, and memory[26]. Therefore, depression may be not only associated with physical frailty but also with cognitive impairment. Multiple studies have confirmed that individuals with depression are at a higher risk of cognitive decline[27–29]. Additionally, depression often co-occurs with sarcopenia[16,30], suggesting a bidirectional relationship that could potentially amplify frailty risk. Moreover, based on the previous studies, sarcopenia causes brain atrophy, which leads to cognitive deterioration. However, cognitive impair-

ment can worsen sarcopenia, producing a vicious cycle, further complicating the relationship between these conditions. Given these observations, it is plausible to hypothesize that depression and cognitive function may act as mediating factors in the relationship between sarcopenia and frailty. By investigating the relationships between sarcopenia, depression, cognitive function, and frailty, this study aims to uncover the underlying pathways through which these factors interact. Understanding these pathways is critical for developing targeted interventions to prevent or mitigate frailty in older adults. The findings of this study could inform the creation of comprehensive care strategies that address not only the physical aspects of sarcopenia but also the psychological and cognitive function that contribute to frailty. This holistic approach is essential for improving the quality of life and functional status of older adults at risk of frailty. Consequently, the aims of this investigation were twofold: (1) to investigate the relationships between sarcopenia, depression, cognitive function, and frailty, and (2) to explore the possible pathways from sarcopenia, depression, and cognitive function to frailty.

Methods

Data source

The data utilized in our research was derived from the China Health and Retirement Longitudinal Study (CHARLS). Additional information concerning the recruitment strategies and sampling methods employed in CHARLS is available in earlier publications [31]. The data from the 2011–2015 were employed in this research. The criteria for exclusion were defined in the following manner: (1) participants who did not undergo an assessment for frailty in Wave 1 (2011); (2) those identified as frail in Wave 1; (3) subjects

who were not successfully followed up for frailty measurement in Wave 2 (2013); and (4) participants who did not have successful follow-up frailty measurements in Wave 3 (2015). The analysis included a total of 3,108 participants who were 45 years of age or older, selected according to the predefined inclusion criteria. A thorough overview of the participant selection process is illustrated in Fig. 1.

Ethics statement

This investigation was carried out in accordance with all applicable ethical standards. The Institutional Review Board of Peking University (IRB00001052-11015) conducted a thorough review and granted approval for the study. Prior to their participation, each individual provided informed consent, ensuring their willingness to be included in the research.

Measures

Sarcopenia

The evaluation of sarcopenia status was conducted via the AWGS2019 algorithm, which consists of three key components: muscle strength, appendicular skeletal muscle mass (ASM), and physical performance[32]. Muscle strength was quantified through a grip strength assessment employing a Yuejian TM WL-1000 grip dynamometer. During this test, each participant was instructed to hold the dynamometer at a right angle with both hands while performing the assessment. Measurements were taken for both dominant and nondominant hands, with the highest value from two attempts recorded. If the measurement of one hand could not be performed for certain reasons, the highest value obtained from the alternate hand was documented. As per the AWGS 2019 guide-

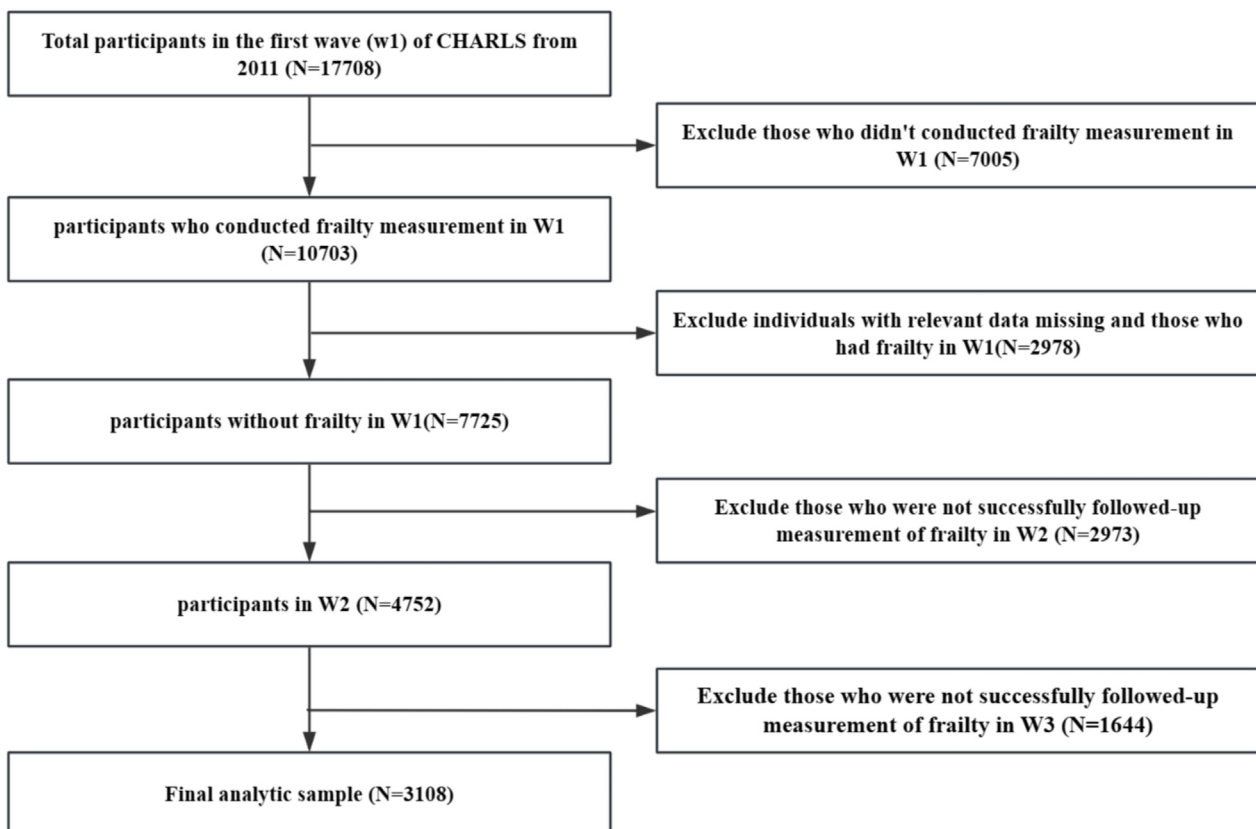


Fig. 1. Flow diagram for participants included in the study.

lines, the threshold for low grip strength was set at under 28 kg for males and under 18 kg for females. The ASM for Chinese residents was estimated via validated anthropometric equations[33]:

$$\text{ASM} = 0.193 \times \text{weight}(\text{kg}) + 0.107 \times \text{height}(\text{cm}) - 4.157 \times \text{sex} - 0.037 \times \text{age}(\text{years}) - 2.631$$

The ASM derived from the formula closely matched the outcomes obtained through dual-energy X-ray absorptiometry (DXA) [34]. Consistent with prior studies[35,36], the threshold for identifying low muscle mass was established by evaluating the lowest 20 % of sex-specific height-adjusted muscle mass (ASM/Ht²) within the population under investigation. Within the framework of our study, ASM/Ht² values that fell below 5.29 kg/m² for females and below 7.01 kg/m² for males were deemed indicative of low muscle mass. To evaluate physical performance, the five-time chair stand test was employed, wherein we measured the duration, expressed in seconds, that participants took to rise from a chair with their arms crossed and complete five consecutive times. In the analysis, patients who attempted the chair stand test but were unable to finish it were categorized as having low physical performance. As per the AWGS 2019 guidelines, a duration of 12 s or more in the five-time chair stand test was used as the threshold to classify low physical performance.

In this research, the identification of potential sarcopenia was based on the observation of diminished muscle strength, which may occur either in the absence of or alongside reduced physical performance. The definition of sarcopenia is defined to include not only a decrease in muscle mass but also the coexistence of low muscle strength or impaired physical performance. Participants who exhibited no manifestations of weak muscle strength, diminished muscle mass, or poor physical performance were designated “nonsarcopenia”[37]. Consequently, the entire group of participants was segmented into three distinct categories: nonsarcopenia, possible sarcopenia, and sarcopenia.

Depression

The Centre for Epidemiological Studies Depression Scale (CESD-10) was utilized to evaluate depression, demonstrating strong validity and favourable psychometric properties among elderly adults in China[38]. This scale serves as a streamlined self-report tool for assessing depressive symptoms, aimed at gauging an individual's depressive experiences over the past week. The CESD-10 is comprised of 10 individual items, each with four response options representing varying frequencies (from “rarely” or “none of the time” to “most” or “all of the time”). Items 5 and 8 are scored in reverse, contributing to the overall assessment. The total score can range from 0 to 30, where elevated scores indicate a greater severity of depressive symptoms.

Cognitive function

To assess cognitive function, the approach utilized in the American Health and Retirement Study (HRS) was adopted, where the applied methodology formed the basis of evaluation[39]. The assessment extended across multiple facets of cognitive abilities, specifically covering four distinct domains: orientation, memory, calculation, and drawing. The orientation dimension included five items (year, month, day, day of the week, and season), with each assigned a value of one point, for a total of five points. For calculation, participants were instructed to subtract 7 from 100 in a continuous sequence five times, and each successful subtraction earning one point. Immediate memory was assessed by reading ten random words to each participant and counting the number

of words they could recall at once. The assessment of delayed recall took place following the subjects' completion of the depression scale survey, calculation, and drawing tests. The memory dimension score was determined by combining the results from both the immediate recall and delayed recall tasks, with each recalled word earning one point, for a total of 20 points. Finally, participants were presented with an image of two overlapping pentagrams and were tasked with creating an appropriate pentagram to assess their drawing skills, with one point granted for each accurate representation. The overall score for cognitive function varied from 0 to 31, where higher scores reflected superior cognitive performance.

Frailty

Frailty was evaluated through the Frailty Index (FI), a metric derived from the aggregation of various health deficits associated with aging. In this study, the FI was meticulously developed in accordance with the standardized procedures that have been previously outlined, where an index containing 30–40 variables have been shown to be sufficient for accurately predicting adverse outcomes[40]. Therefore, this study selected 36 items to construct the FI, encompassing aspects of disease, disability, activities of daily living, and mobility. The 36 health variables were assigned values of 0 or 1 (where 0 signifies the absence of a health deficit and 1 indicates the presence of a health deficit). The FI derives from the calculation of the ratio of the total number of current health deficits to 36, with a score ranging from 0 to 1. According to previous research, an FI ≥ 0.25 was defined as frailty[41]. The specific items used to construct the FI are detailed in Table S1.

Covariates

The covariates considered in our research encompassed both demographic characteristics and lifestyle variables. The demographic characteristics consisted of age, sex, educational level, marital status, and current hukou status, whereas the lifestyle variables encompassed smoking habits, alcohol consumption, and sleep duration. In alignment with findings from earlier research [42], sleep duration was classified among three distinct groups: short sleep duration (6h or fewer per night), normal sleep duration (between 7h and 9h per night), and long sleep duration (10h or more per night). The coding details for all the variables are presented in Table S2.

Statistical analysis

In the context of descriptive statistics, continuous variables take the form of means ± standard deviations (SDs) or medians accompanied by interquartile ranges, whereas categorical variables are represented in terms of percentages. The baseline characteristics of the nonfrail and frail study populations were assessed using chi-square tests and t-tests for comparison. Cox regression models served the purpose of analyzing the relationships between sarcopenia, depression, cognitive function, and the incidence of frailty, with hazard ratios (HRs) and their 95 % confidence intervals (CIs). The calculation of the follow-up duration was based on the interview year and month provided by the CHARLS. Within the framework of the multivariate Cox regression model, baseline demographic and lifestyle variables were controlled. In the sensitivity analyses, adjustments were made to account for demographic and lifestyle factors. Interaction effects were analysed via binary logistic regression, revealing an impactful interaction between sarcopenia and depression/cognitive function in terms of frailty status (Table S3). Consequently, stratified analyses were performed according to sarcopenia categories, and associations

between depression, cognitive function, and frailty status within different sarcopenia strata were estimated. Structural equation modelling was utilized for mediation effect analysis to explore possible pathways from sarcopenia, depression, and cognitive function to frailty status. IBM SPSS (Version 25.0) and the PROCESS v4.1 plugin were used, selecting Model 6 where the bootstrap sample count was established at 5,000. The presence of a significant mediation effect was considered when the 95 % CI failed to include the value of 0 [43]. All analyses were two-sided, with p-values that fell below 0.05 being regarded as statistically significant.

Results

Participant characteristics

Table 1 provides an overview of the baseline characteristics of the nonfrail and frail study populations. The average age of the 3,108 participants was 60.15 (SD = 9.07) years, with 1,937 (62.3 %) females. Throughout the course of the 4-year follow-up duration, 753 participants developed frailty (294 cases of newly occurring frailty at Wave 2 and 459 cases at Wave 3).

Associations between sarcopenia, depression, cognitive function, and frailty

In the Cox regression analysis, we estimated three distinct models: Model 1 was the crude model, Model 2 controlled for demographic characteristics, and Model 3 further controlled for lifestyle variables on the basis of Model 2. The data presented in Table 2 demonstrate that, sarcopenia, depression, and cognitive function were found to be significantly related to the onset of

frailty. These associations remained robust despite the adjustments made for demographic characteristics and lifestyle variables.

Associations between depression, cognitive function, and frailty stratified by sarcopenia status

Table 3 presents the associations between depression, cognitive function, and frailty categorized according to. In the cohort of individuals classified as nonsarcopenia and possible sarcopenia at baseline, depression [HR (95 % CI): 1.034 (1.017, 1.051)], and cognitive function [HR (95 % CI): 0.975 (0.955, 0.995)] were significantly and independently linked to the development of frailty. In contrast, among those diagnosed with sarcopenia at baseline, depression remained significantly and independently linked to the onset of frailty [HR (95 % CI): 1.051 (1.018, 1.085)], whereas cognitive function failed to demonstrate a significant association with the onset of frailty.

Pathways from sarcopenia status, depression, and cognitive function to frailty

In this study, the independent variable is a multicategorical variable. Following the method proposed by Hayes et al.[44], we created two dummy variables, D1 (possible sarcopenia) and D2 (sarcopenia), with nonsarcopenia as the control group. Table 4 and Fig. 2 illustrate an illustration of the relative direct and indirect effects associated with these two dummy variables, with the overall relative effect being defined as the aggregate of the relative direct, as well as the indirect effects. Relative to the nonsarcopenia group, both the relative direct and indirect effects of possible sar-

Table 1
Baseline characteristics of participants grouped by frailty status.

Characteristics	Total (N = 3108)	Nonfrail group (N = 2355)	Frail group (N = 753)	P value	
Main variables					
Classification of sarcopenia				< 0.001	
No sarcopenia	1955 (62.9)	1554 (79.5)	401 (20.5)		
Possible sarcopenia	873 (28.1)	617 (70.7)	256 (29.3)		
sarcopenia	280 (9.0)	184 (65.7)	96 (34.3)	< 0.001	
Depression, median (IQR)	9 (5, 14)	8 (4, 13)	11 (6, 15)		
Cognitive function, median (IQR)	13 (10, 17)	14 (10, 18)	12 (9, 16)		
Covariates					
Demographic characteristics					
Age (Mean ± SD)	60.15 ± 9.07	59.41 ± 8.96	62.48 ± 9.02	< 0.001	
Sex				0.149	
Female	1937 (62.3)	1451 (74.9)	486 (25.1)	0.238	
Male	1171 (37.7)	904 (77.2)	267 (22.8)		
Marital status					
Single	534 (17.2)	394 (73.8)	140 (26.2)	< 0.001	
Partnered	2574 (82.8)	1961 (76.2)	613 (23.8)		
Educational level					
Primary school or less	2326 (74.8)	1715 (73.7)	611 (26.3)	0.14	
Middle school	557 (17.9)	461 (82.8)	96 (17.2)		
High school or above	225 (7.2)	179 (79.6)	46 (20.4)		
Current Hukou status				0.462	
Agricultural Hukou	2583 (83.1)	1944 (75.3)	639 (24.7)		
Non-Agricultural Hukou	525 (16.9)	411 (78.3)	114 (21.7)		
Lifestyle characteristics				0.063	
Smoking					
No	2050 (66.0)	1545 (75.4)	505 (24.6)		
Yes	1058 (34.0)	810 (76.6)	248 (23.4)	0.006	
Alcohol drinking					
No	2284 (73.5)	1711 (74.9)	573 (25.1)		
Yes	824 (26.5)	644 (78.2)	180 (21.8)		
Sleep duration				0.006	
Short sleep	1721 (55.4)	1267 (73.6)	454 (26.4)		
Normal sleep	1268 (40.8)	998 (78.7)	270 (21.3)		
Long sleep	119 (3.8)	90 (75.6)	29 (24.4)		

Abbreviations: SD, standard deviation.

Table 2

The associations between sarcopenia, depression, cognitive function, and frailty status.

Variables	Model 1 ^I		Model 2 ^{II}		Model 3 ^{III}	
	HR (95 %CI)	P value	HR (95 %CI)	P value	HR (95 %CI)	P value
No sarcopenia						
Possible sarcopenia	1.372 (1.172, 1.606)	***	1.363 (1.164, 1.596)	***	1.354 (1.156, 1.586)	**
sarcopenia	1.560 (1.241, 1.961)	***	1.537 (1.222, 1.934)	***	1.514 (1.203, 1.907)	***
Depression	1.036 (1.024, 1.047)	***	1.035 (1.023, 1.046)	***	1.033 (1.021, 1.044)	***
Cognitive function	0.973 (0.960, 0.986)	***	0.977 (0.963, 0.991)	***	0.977 (0.963, 0.991)	***

Note: HR, Hazard ratios.

^I Model 1 was crude model.

^{II} Model 2 was adjusted for demographic characteristics.

^{III} Model 3 was adjusted as model 2 with further adjustment for lifestyle characteristics.

** P < 0.01.

*** P < 0.001.

Table 3

The relationships among depression, cognitive function, and frailty status stratified by sarcopenia status.

Variables	Sarcopenia status		
	No sarcopenia HR (95 %CI)	Possible sarcopenia HR (95 %CI)	Sarcopenia HR (95 %CI)
Depression	1.034 (1.017, 1.051)***	1.024 (1.006, 1.043)*	1.051 (1.018, 1.085)**
Cognitive function	0.975 (0.955, 0.995)*	0.971 (0.947, 0.995)*	1.011 (0.971, 1.054)

Note: Multivariate Cox proportional hazards model adjusted for demographics and lifestyle characteristics.

* p < 0.05.

** P < 0.01.

*** p < 0.001.

Table 4

The pathways from dummy variables (D1and D2), depression and cognitive function to frailty status.

Pathways	Path coefficient (β)	Bootstrap95% CI
Relative direct effect		
Sarcopenia	Reference	Reference
D1 → Frailty	0.4108*	(0.2187, 0.5934)
D2 → Frailty	0.5638*	(0.2666, 0.8359)
Relative indirect effect		
D1 → Depression → Frailty	0.0457*	(0.0204, 0.0757)
D1 → Cognitive function → Frailty	0.0242*	(0.0083, 0.0468)
D1 → Depression → Cognitive function → Frailty	(0.0015, 0.0082)	
D2 → Depression → Frailty	0.0458*	(0.0075, 0.0898)
D2 → Cognitive function → Frailty	0.1066*	(0.0497, 0.1703)
D2 → Depression → Cognitive function → Frailty	0.0044*	(0.0006, 0.0094)

* P < 0.05.

copenia on frailty were significant (the 95 % CI of each effect did not include 0). The relative direct effect of possible sarcopenia on frailty [$\beta = 0.4108$, 95 % CI (0.2178, 0.5934)] constituted 83.4 % of the relative total effect. The relative mediation effects of depression and cognitive function on the total relative effect of possible sarcopenia on frailty were 9.3 % and 4.9 %, respectively, with an effect value of 0.0044 through depression and cognitive function, accounting for 0.9 % of the total effect. Similarly, both the relative direct and indirect effects of sarcopenia on frailty were significant relative to those of the nonsarcopenia group. The relative direct

effect of sarcopenia on frailty [$\beta = 0.5638$, 95 % CI (0.2666, 0.8359)] The relative direct effect of sarcopenia on frailty [$\beta = 0.5638$, 95 % CI (0.2666, 0.8359)] constituted 78.2 % of the relative total effect. 78.2 % of the relative total effect. The relative mediation effects of depression and cognitive function on the total relative effect of sarcopenia on frailty were 6.3 % and 14.7 %, respectively, with an effect value of 0.0044 through depression and cognitive function, accounting for 0.6 % of the total effect.

Discussion

This research represents the pioneering prospective investigation into the correlations among sarcopenia, depression, cognitive function, and the occurrence of frailty in middle-aged and older individuals. Through the use of a sample that is nationally representative, the research yielded several notable findings. First, baseline sarcopenia, depression, and cognitive function were significantly connected to the onset of new-onset frailty. Second, the relationships among depression, cognitive function, and frailty differ depending on the various statuses of sarcopenia. Third, path analysis showed that relative to nonsarcopenia, possible sarcopenia/sarcopenia was associated with frailty through depression and cognitive function, respectively, and sequentially through depression and cognitive function.

The findings presented in this research demonstrate that after adjusting for various demographic and lifestyle factors, sarcopenia is significantly linked to frailty, aligning with the findings of a prior study[45]. Although sarcopenia and frailty are classified as distinct conditions, they share many similarities in aetiology and symptomatology[46], as do common metabolic, haematological, and inflammatory biomarkers[47]. Consequently, sarcopenia is closely linked to frailty and may exert a regulatory influence over the transition to a state of increased frailty[48]. These results align with the “frailty cycle” theory proposed by Fried et al., which posits that sarcopenia, characterized by muscle loss and altered physical function, may be a pivotal factor in both the initiation and advancement of frailty[3]. Furthermore, our findings concerning

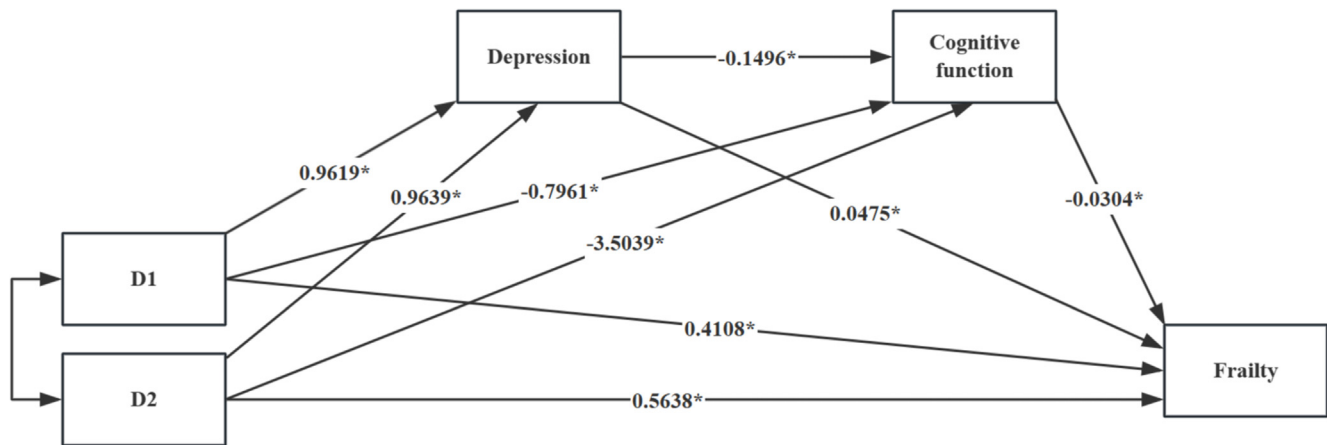


Fig. 2. The path coefficient (β) from dummy variables, depression and cognitive function to frailty status.

the relationships among depression, cognitive function, and frailty suggest that the impacts of depression and cognitive function on frailty vary across different statuses of sarcopenia. Depression has a significant effect on frailty across all groups (nonsarcopenia, possible sarcopenia, and sarcopenia), with multiple studies indicating that baseline depression may precipitate adverse health outcomes during follow-up, including frailty[49]. This association may be due to shared pathological pathways, such as inflammation, associated with both depression and frailty[50,51]. The protective effect of cognitive function against frailty, however, is observed only in nonsarcopenic and possibly sarcopenic individuals, with no significant protective effect in sarcopenic individuals. Research by Panza et al.[52] suggested that a heightened risk of functional disability in older individuals is linked to cognitive decline, which in turn accelerates frailty. Thus, maintaining good cognitive function may help preserve physical function and slow the progression of frailty. The reason for the lack of a notable protective influence on cognitive function among individuals with sarcopenia remains speculative: the decline in muscle function may be the predominant factor influencing frailty status in this group, potentially diminishing the relative impact of cognitive function. The mechanisms underlying this phenomenon warrant further investigation.

Path analysis revealed that depression, cognitive function, and their serial mediation partially clarify the relationship between sarcopenia and frailty. This finding extends beyond previous studies by providing a comprehensive framework that integrates multiple pathways linking sarcopenia to frailty. While earlier research has established a higher prevalence of depression among sarcopenia individuals, with an overall occurrence rate of 28 % [30], and Chen's cohort study confirmed sarcopenia as a distinct independent risk factor for depression[53], our study uniquely explores the serial mediation role of depression and cognitive function in this relationship. Furthermore, our findings align with and expand upon the work of Arts et al., who suggested that depression is linked to both frailty and cognitive impairment, which may persist even during periods of emotional remission[54]. However, our study goes a step further by demonstrating how cognitive dysfunction, often associated with depression, serves as a critical factor in the development of frailty. This connection is not only explained by overlapping mechanisms such as chronic inflammation, oxidative stress, mitochondrial dysfunction, and dysregulation of the hypothalamic–pituitary–adrenal (HPA) axis[49], but also by the serial mediation model we propose, which highlights the sequential interplay between sarcopenia, depression, and cognitive function. In summary, our study significantly advances the current understanding of the complex relationships among sarcopenia,

depression, cognitive function, and frailty. By uncovering these serial mediation pathways, we provide novel insights into the underlying mechanisms driving these interconnections. This not only reinforces the importance of addressing depression and cognitive function in sarcopenia management but also offers a theoretical foundation for early identification, targeted intervention, and continuous monitoring of vulnerable middle-aged and elderly Chinese populations. Such approaches hold the potential to reduce frailty incidence and improve quality of life, thereby filling a critical gap in the current evidence base.

Limitations

There are still several noteworthy limitations in the current study. To begin with, although a wide range of variables influencing depression, cognitive function, and frailty were incorporated into the model, certain additional confounding factors, such as nutrition, exercise, and social participation, were neither measured nor adjusted for during the analysis, which could potentially introduce bias. Additionally, alcohol use (and misuse) is highly important covariate, which may impact cognition, and be linked with depression symptoms[55], the study's binary classification of participants based solely on alcohol consumption, without differentiating between occasional drinkers, heavy drinkers, or those with alcohol misuse or addiction issues, may have overlooked the varying degrees of impact that different levels of alcohol consumption can have on cognition and depression symptoms. This simplification could obscure more detailed associations between specific patterns of alcohol use and cognitive outcomes, and it precludes the exploration of potential moderating or mediating effects of factors such as frequency of drinking, quantity consumed, or the presence of alcohol dependence, which are crucial for understanding the full spectrum of alcohol's effects on cognitive health. Furthermore, both the assessment of depressive symptoms and the construction of the frailty index in this research were reliant on self-reported data, suggesting that future studies should consider employing more objective and standardized measurement techniques. Finally, missing values were excluded from the present study, and no imputation methods—despite their effectiveness—were utilized, potentially contributing to some degree of bias into the outcomes.

Conclusion

In summary, we found that sarcopenia, depression, cognitive function, and frailty are interrelated; sarcopenia is not only associ-

ated with frailty through depression and cognitive function separately but also affects frailty via the chain-mediated roles of both depression and cognitive function in tandem. The varying impact exerted by depression and cognitive function on frailty varies across different statuses of sarcopenia, which could assist in shaping more specifically tailored strategies. For nonsarcopenia, possible sarcopenia, and sarcopenia, comprehensive therapies aimed at promoting mental health and alleviating depressive symptoms are critical for preventing frailty. While the assessment and improvement of cognitive function may be of greater significance in non-sarcopenia and possible sarcopenia, we advocate for a holistic approach that integrates interventions for sarcopenia or mental health issues with measures to slow cognitive decline to prevent frailty.

Compliance with ethics requirements

This study was conducted in compliance with all relevant ethical standards. The study was reviewed and approved by the Institutional Review Board of Peking University (IRB00001052-11015). Each participant provided informed consent prior to their inclusion in the study.

Funding

This work was supported by the Health Commission of Sichuan Province (grant numbers 2023YFS0070). The funders played no role in the study design, data collection and analysis, publication decision, or manuscript preparation.

Data availability statement

For access to the original data, please refer to the following link: <https://charls.pku.edu.cn/>.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

We are grateful to all volunteers and staff who participated in China Health and Retirement Longitudinal Study for collecting and providing data.

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jare.2024.12.021>.

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